BAI-2A

TASK 03

K25-0064
SARIM RAZA ANSARI

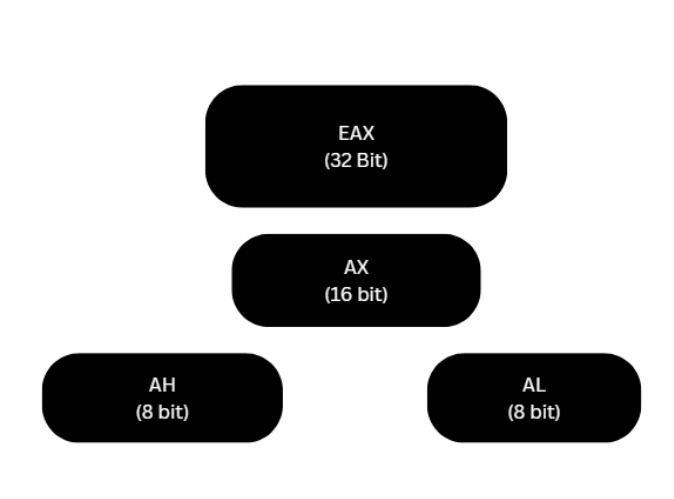
Task 01:

EAX: A 32 bit register used to perform arithmetic operations and storing values.

EBX: A 32 bit register use to hold values.

ECX: A 32 bit register used as counter.

EDX: A 32 bit register use to hold data and perform input/output operations.



ECX: It is commonly use in counters.

EDX: Use in input/output operations.

Task 02:

```
.data

rollNo      WORD 2201              ; Stores the student's roll number
marks       BYTE 87                ; Stores the student's marks out of 100
studentName BYTE "Ali Raza", 0     ; Stores the student's name, null-terminated
grade       BYTE ?                 ; Uninitialized byte for the student's grade
```

Task 03:

```
.data

; Attendance of students from Monday to Friday
attendance BYTE 30, 28, 32, 25, 29

; Five empty/uninitialized bytes for future seat information
emptySeats BYTE 5 DUP(?)
```

Task 04:

```
.data

buffer    BYTE 15 DUP(0)       ; Reserves 15 bytes initialized to 0
tempSpace BYTE 10 DUP(?)       ; Reserves 10 uninitialized bytes
pattern   BYTE 5 DUP("HI")      ; Stores "HIHIHIHIHI"
```

Task 05:

```
MOV CX, 'B'       ; Character constant: ASCII 'B' = 42h = 66 decimal
MOV CX, 42h       ; Hexadecimal: 42h = 66 decimal
MOV CX, 66        ; Decimal: 66 = 42h
MOV CX, 01000010b ; Binary: 01000010b = 42h = 66 decimal
```

Task 06:

```
TITLE Tuck Shop Cash Register (CashRegister.asm)

INCLUDE Irvine32.inc

.code

main PROC

    ; Load opening cash balance into EAX
    MOV EAX, 5000

    ; Add sales income
    ADD EAX, 1500

    ; Subtract supply expenses
    SUB EAX, 800

    ; Display final register values
    CALL DumpRegs

    ; Exit program
    exit

main ENDP
END main
```